

Next Generation Sequencing Data – an Introduction

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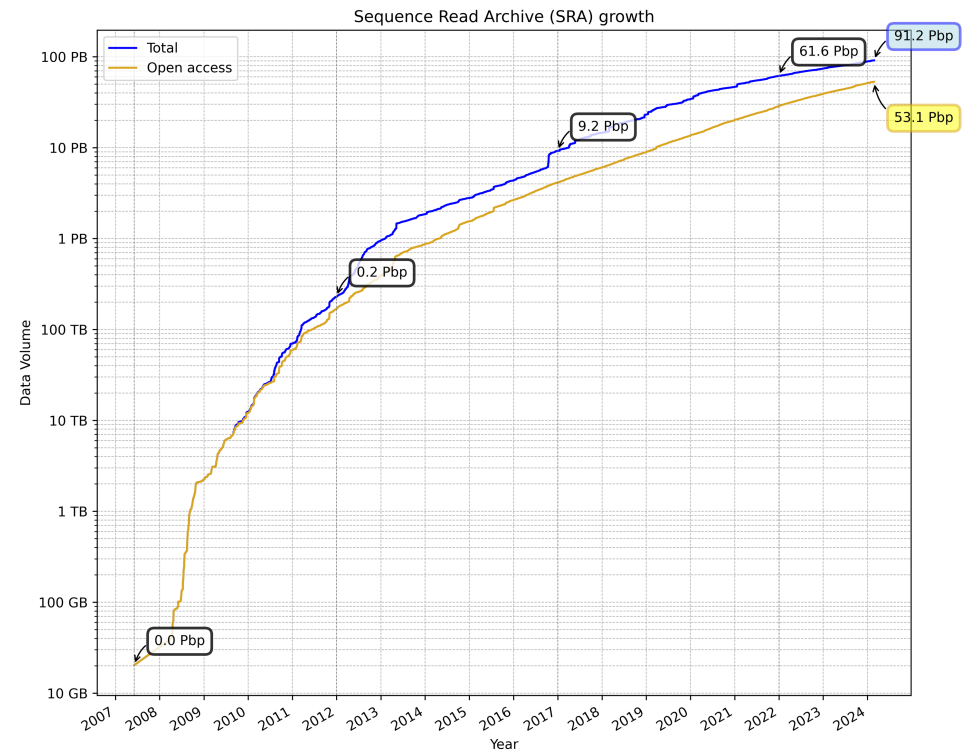
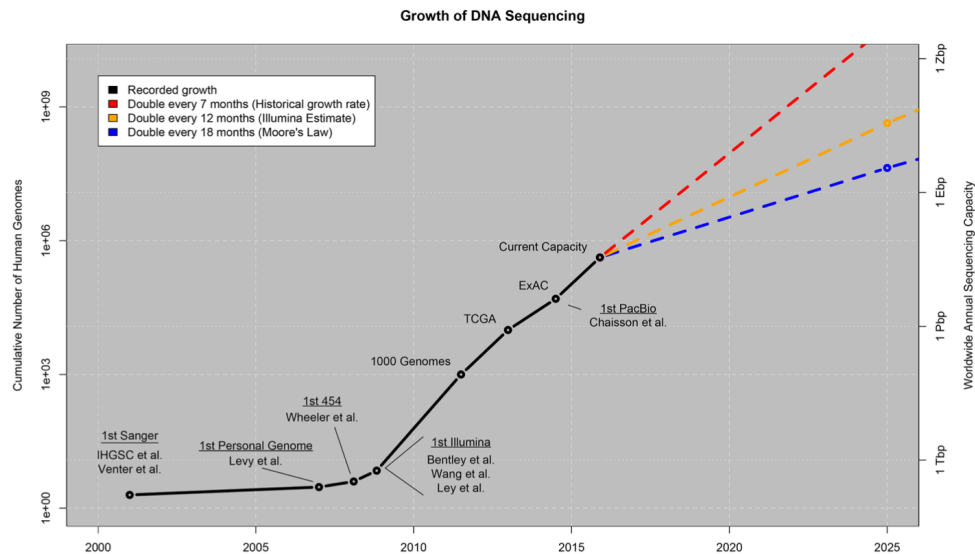


University of
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ETH

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Swiss Federal Institute of Technology Zurich

NGS Data Increase



- NGS data increases exponentially
- sequencing capacity installed (2025): 3×10^{18} bases/year

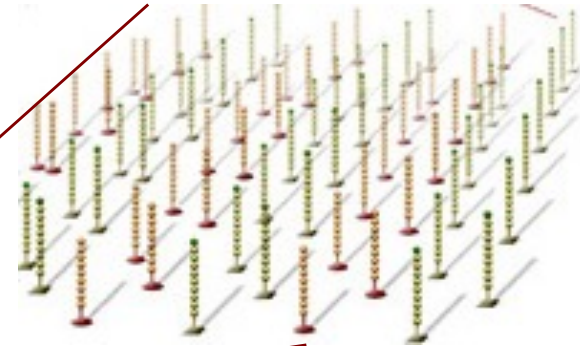
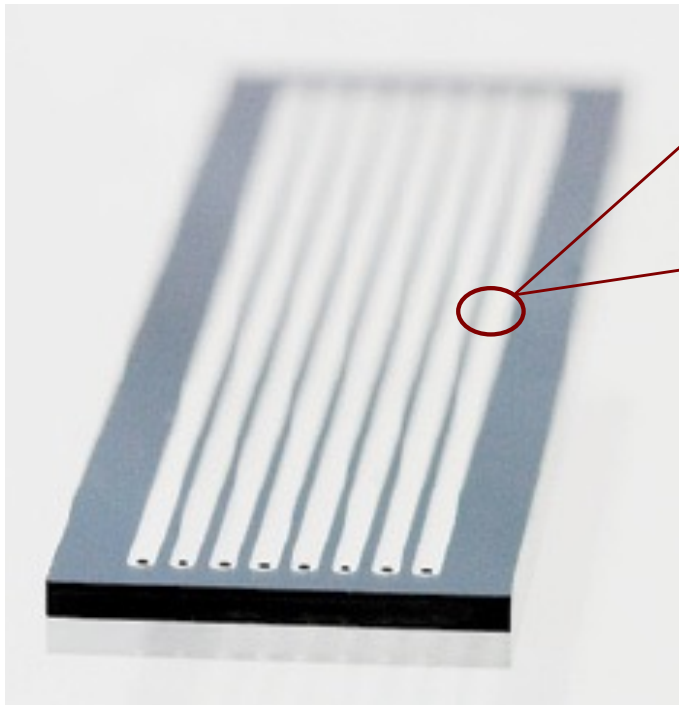
Ingredients for the success

- Evolution has yielded DNA and RNA molecules for information storage and transfer. They have good properties to be read (**measured**)
- NGS technologies rely on
 - processing of molecules is **massively parallel**
 - measurement process is done by individual molecules (**cheap and fast**)
 - actual readout is **massively parallel, cheap and fast**:
 - fluorescent imaging
 - electrical signal in pores or wells

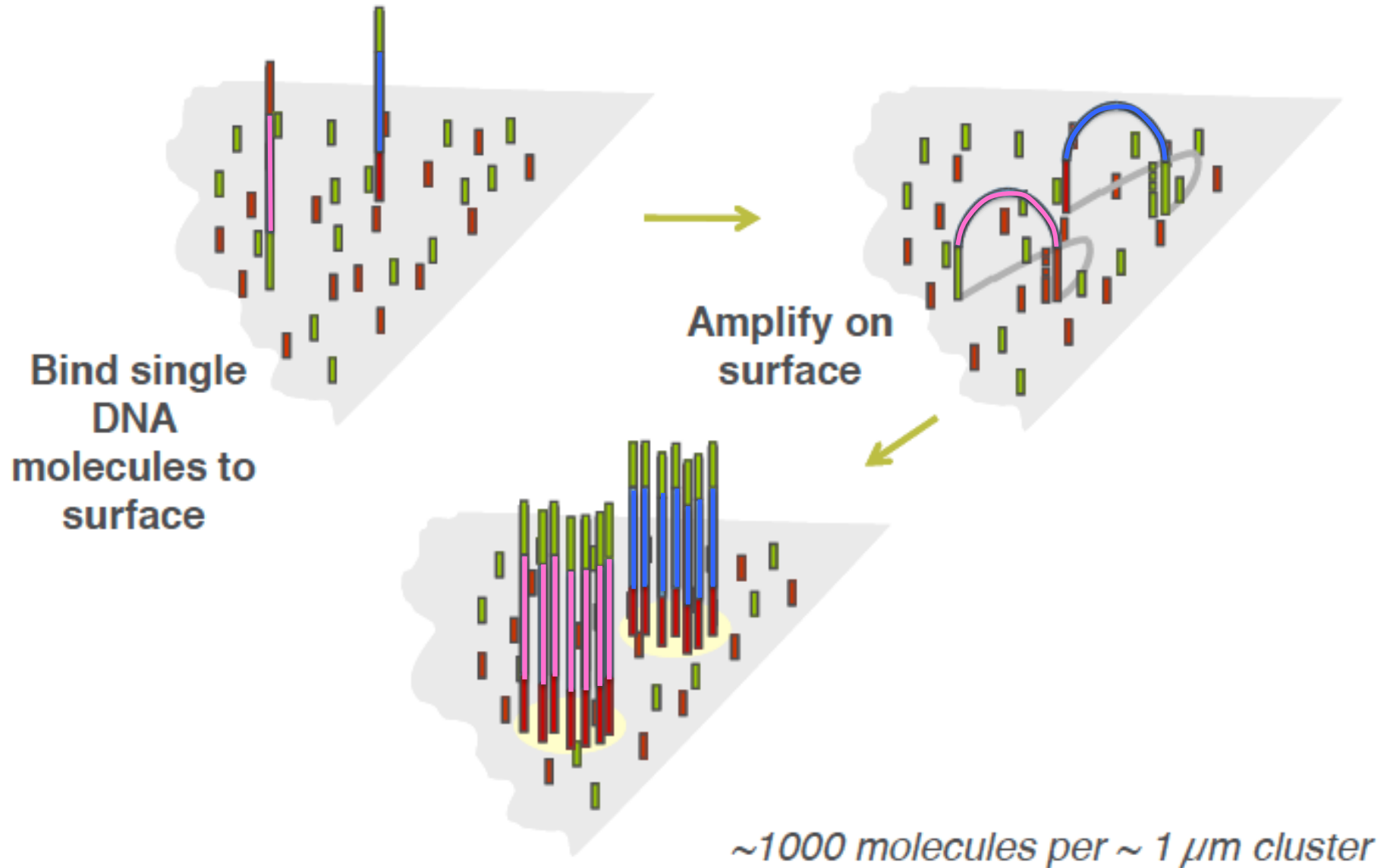
Sequencers

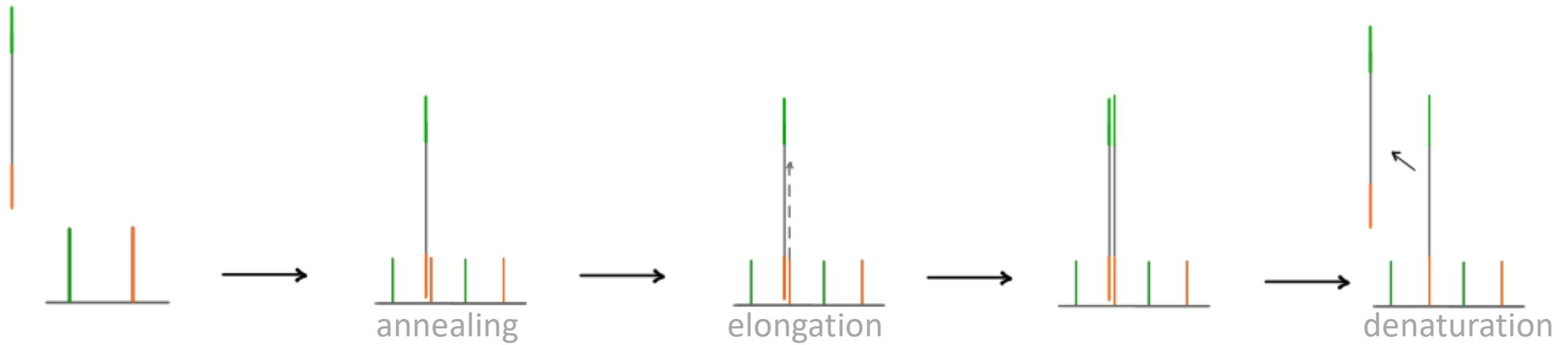
- short-read sequencers (read length: 2×150 bp in routine use, up to 2×300 bp)
 - illumina (e.g. NovaSeq X)
 - Element Biosciences, Aviti
 - Complete Genomics, DNBSEQ-T20X2
- long-read, single molecule technologies (500bp – megabases)
 - PacBio, Revio
 - Oxford NanoPore Technologies, PromethION
- example overview:
 - <https://genohub.com/ngs-instrument-guide/>

Illumina Flow cell

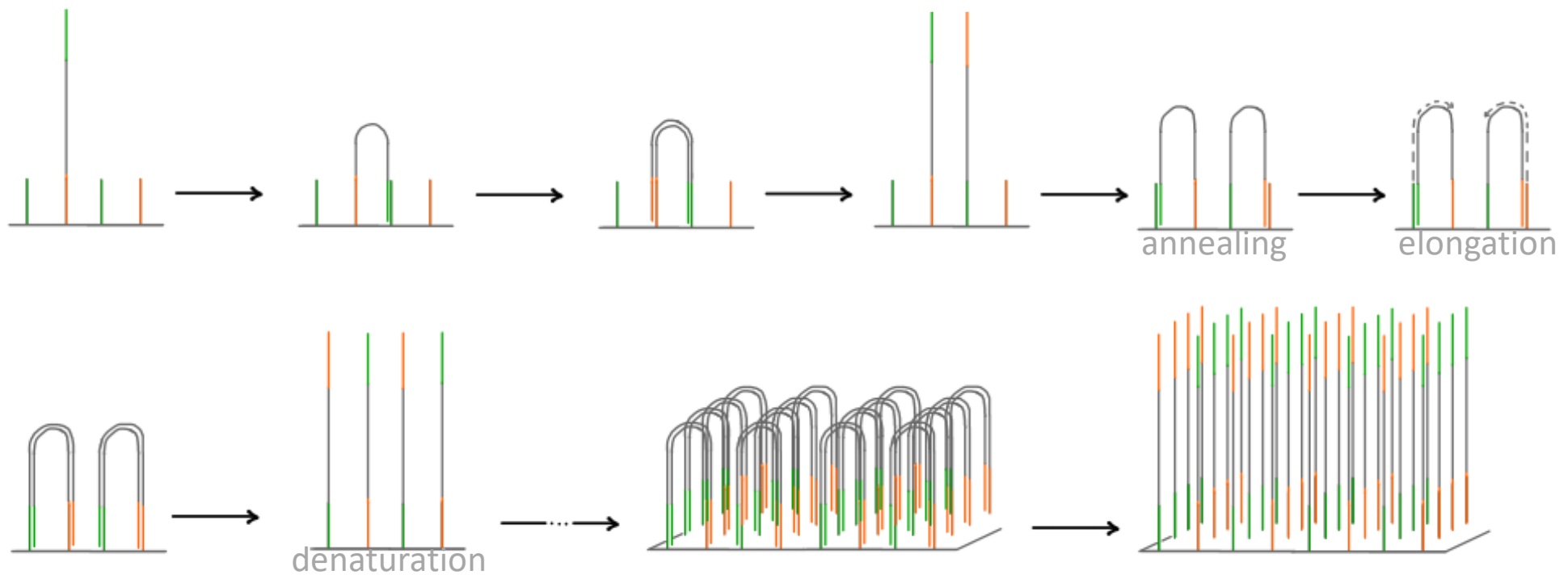


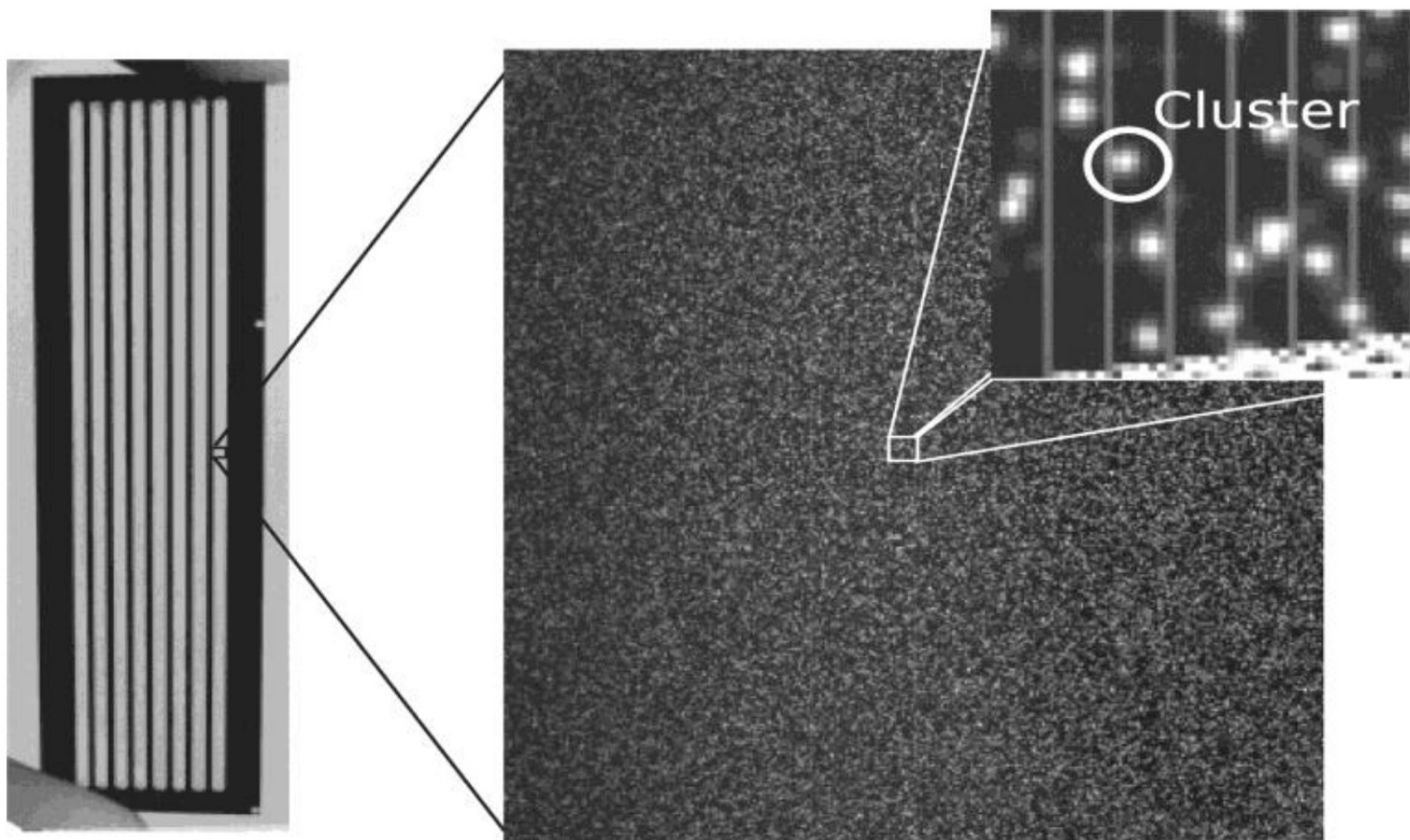
Cluster generation overview



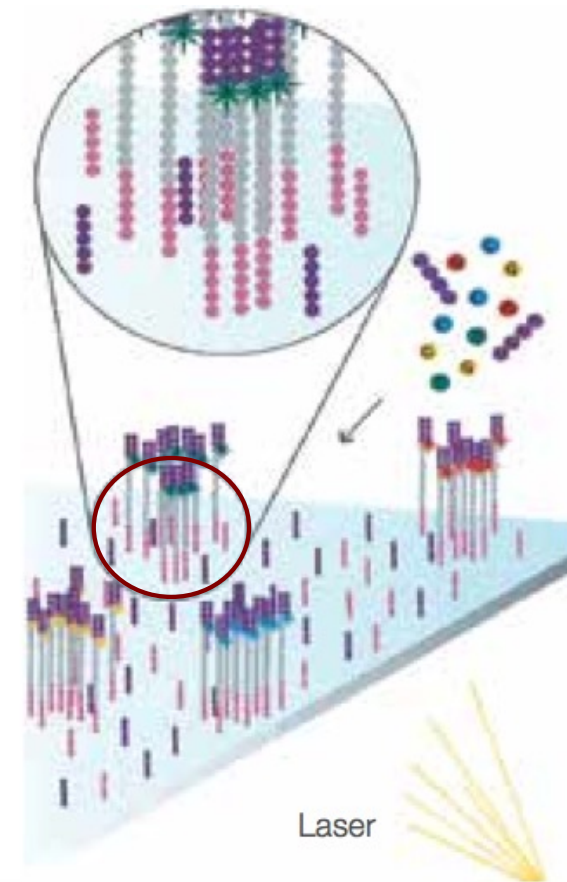


bridge amplification



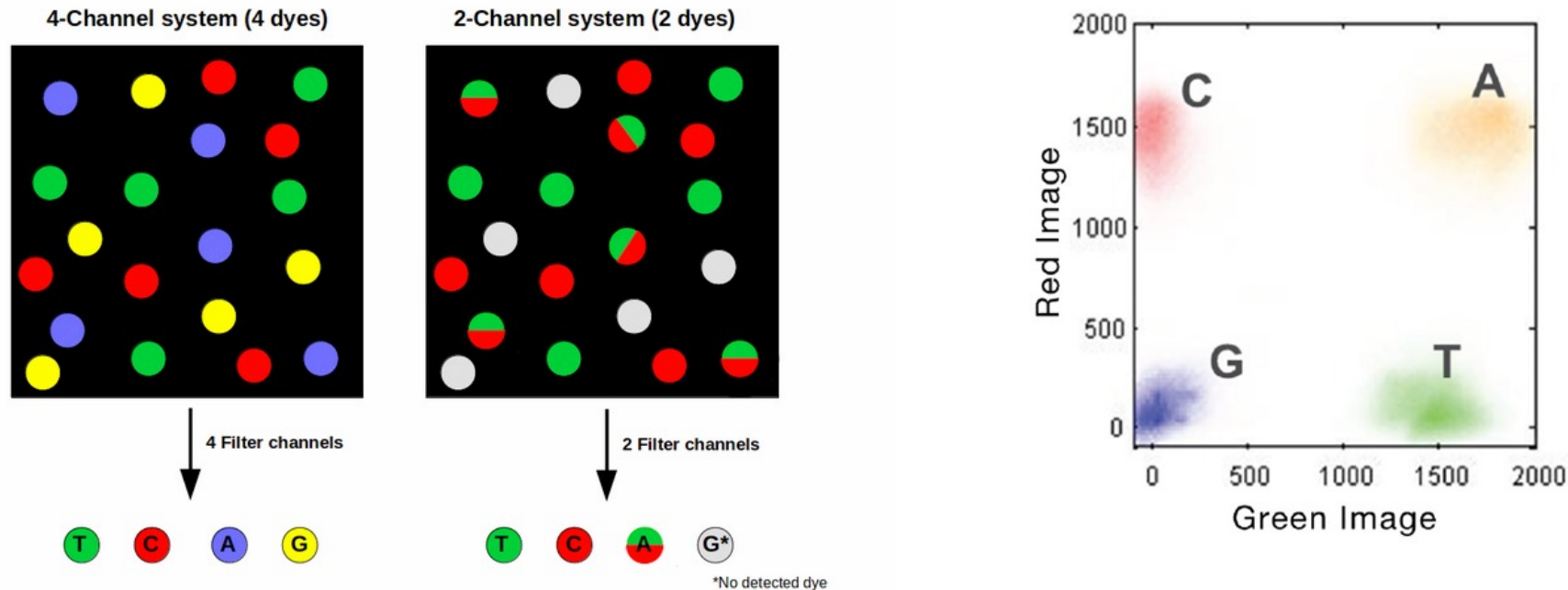


Illumina Sequencing



The first sequencing cycle begins by adding four labeled reversible terminators, primers, and DNA polymerase.

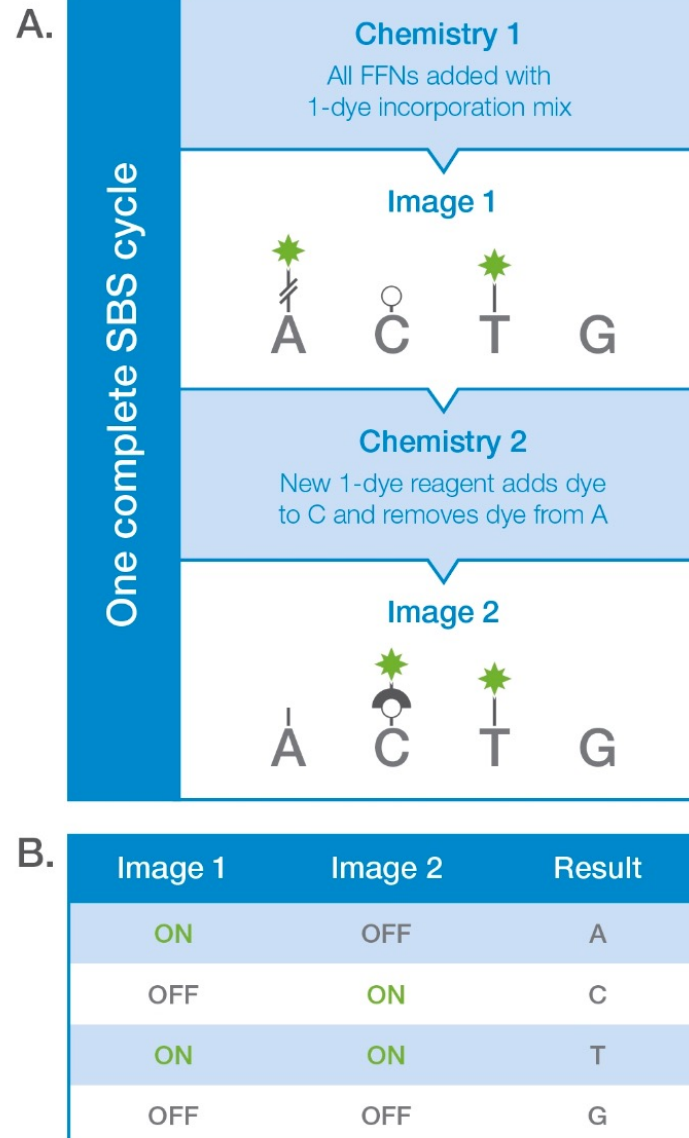
Color coding of bases



- 4-channel: Takes longer, readout is more “expensive”; can identify unknown bases (Ns) as lack of signal
- 2-channel: Faster, cheaper, but can not discriminate between failure to read (N) and a G

Color coding of illumina iSeq

- Single color system



B.

Image 1	Image 2	Result
ON	OFF	A
OFF	ON	C
ON	ON	T
OFF	OFF	G

Sequencer are extremely efficient instruments

For the **sampled** molecules, it answers exactly **one** question: what is the sequence?

It does **not** know:

- which organism, tissue, cell or sample the molecule came from
- whether it is DNA, or cDNA copied from RNA
- whether it is a whole molecule or a fragment of one
- whether it was selected on purpose, or came along by accident
- whether it is an original molecule or the 512th PCR copy of one

Every one of those answers was decided **before** the library reached the flow cell.

→ Understanding the sample and library prep is key to subsequent analysis

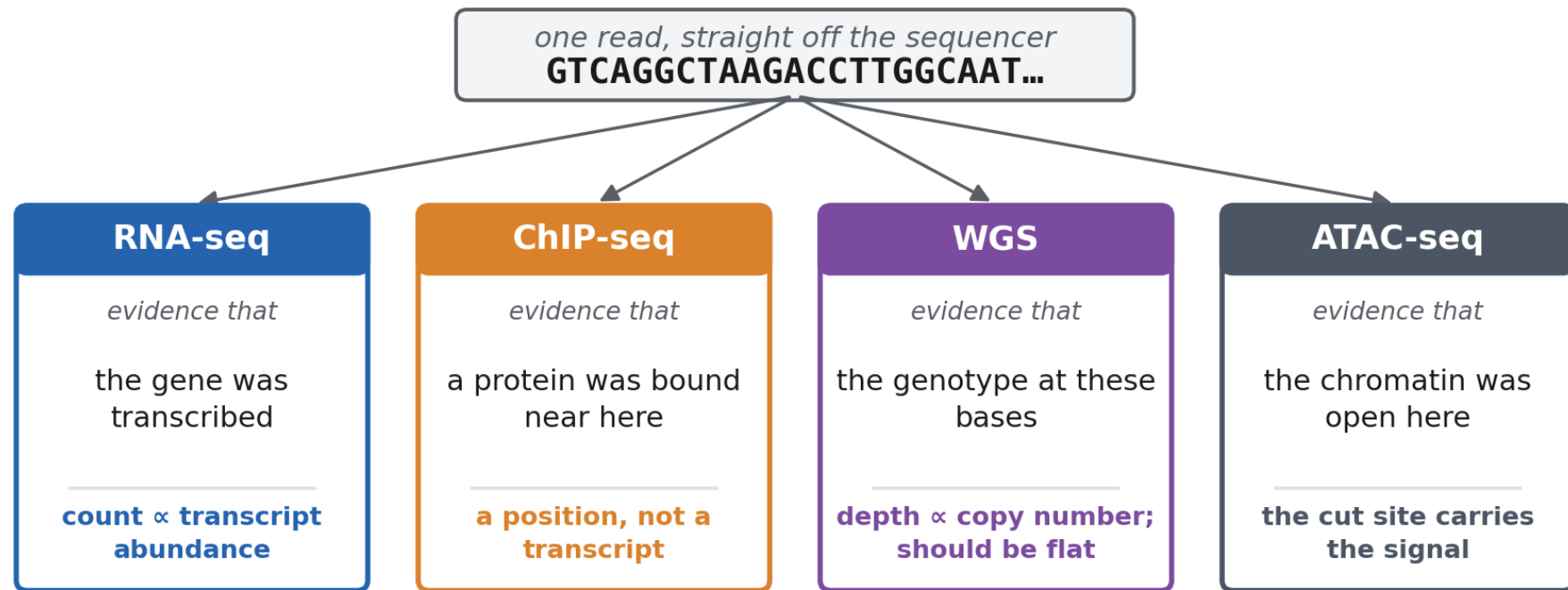
What the Library Prep Decides — and the Sequencer Cannot

Features decided by the Library Prep

- **which molecules are captured** — poly-A selection, rRNA depletion, a capture panel, an antibody, size selection
→ **what your counts are counts of**
- **what gets attached** — adapters, sample index, UMI, cell barcode
→ which read belongs to which sample, cell or place; and whether you can count molecules rather than reads
- **fragmentation and insert size** — how much of each molecule a read can see
→ mappability, junction and isoform resolution

The Same Read, Four Different Meanings

The bases below are identical in all four experiments



- Same data record is generated by the sequencer but downstream analysis workflows and interpretation will differ

Phred scores measure base call accuracy

- P
 - error probability of a given base call
- Q
 - $-10\log_{10}P$
- Assign to each base
- Range from 0-41



Phred Quality Score	Probability of Incorrect Base Call	Base Call Accuracy
10	1 in 10	90%
20	1 in 100	99%
30	1 in 1,000	99.9%
40	1 in 10,000	99.99%
50	1 in 100,000	99.999%

Ewing B, Green P. 1998. Genome Res. 8(3):186-194.

http://en.wikipedia.org/wiki/Phred_quality_score

Phred scores are stored with sequences

- FASTQ

- 4 lines:

1. Header line for Read (starts with “@” and the sequence ID)
2. Sequence
3. Header line for Qualities (starts with “+”)
4. Quality score (error probability; represented in ASCII format)

```
@HWI-ST1034:40:C08PJACXX:2:1101:20681:1994 1:N:0:ATCACG
CTCGNAGACTGGCAACTTGTTCTGGTTTACTGCACCTTCTTTTAAAGGCAGAAAGGC
+
CCCF#2ADHHHGHJJJIJJHIIJJHIIJJJJJJJJJBGGIIJJJJJJJJJJJJJJJJJJJJ
```

Dec	Hx	Oct	Char	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr
0	0	000	NUL (null)	32	20	040	 	Space	64	40	100	@	@	96	60	140	`	`
1	1	001	SOH (start of heading)	33	21	041	!	!	65	41	101	A	A	97	61	141	a	a
2	2	002	STX (start of text)	34	22	042	"	"	66	42	102	B	B	98	62	142	b	b
3	3	003	ETX (end of text)	35	23	043	#	#	67	43	103	C	C	99	63	143	c	c
4	4	004	EOT (end of transmission)	36	24	044	$	\$	68	44	104	D	D	100	64	144	d	d
5	5	005	ENQ (enquiry)	37	25	045	%	%	69	45	105	E	E	101	65	145	e	e
6	6	006	ACK (acknowledge)	38	26	046	&	&	70	46	106	F	F	102	66	146	f	f
7	7	007	BEL (bell)	39	27	047	'	'	71	47	107	G	G	103	67	147	g	g
8	8	010	BS (backspace)	40	28	050	(	(	72	48	110	H	H	104	68	150	h	h
9	9	011	TAB (horizontal tab)	41	29	051	)	)	73	49	111	I	I	105	69	151	i	i
10	A	012	LF (NL line feed, new line)	42	2A	052	*	*	74	4A	112	J	J	106	6A	152	j	j
11	B	013	VT (vertical tab)	43	2B	053	+	+	75	4B	113	K	K	107	6B	153	k	k
12	C	014	FF (NP form feed, new page)	44	2C	054	,	,	76	4C	114	L	L	108	6C	154	l	l
13	D	015	CR (carriage return)	45	2D	055	-	-	77	4D	115	M	M	109	6D	155	m	m
14	E	016	SO (shift out)	46	2E	056	.	.	78	4E	116	N	N	110	6E	156	n	n
15	F	017	SI (shift in)	47	2F	057	/	/	79	4F	117	O	O	111	6F	157	o	o
16	10	020	DLE (data link escape)	48	30	060	0	0	80	50	120	P	P	112	70	160	p	p
17	11	021	DC1 (device control 1)	49	31	061	1	1	81	51	121	Q	Q	113	71	161	q	q
18	12	022	DC2 (device control 2)	50	32	062	2	2	82	52	122	R	R	114	72	162	r	r
19	13	023	DC3 (device control 3)	51	33	063	3	3	83	53	123	S	S	115	73	163	s	s
20	14	024	DC4 (device control 4)	52	34	064	4	4	84	54	124	T	T	116	74	164	t	t
21	15	025	NAK (negative acknowledge)	53	35	065	5	5	85	55	125	U	U	117	75	165	u	u
22	16	026	SYN (synchronous idle)	54	36	066	6	6	86	56	126	V	V	118	76	166	v	v
23	17	027	ETB (end of trans. block)	55	37	067	7	7	87	57	127	W	W	119	77	167	w	w
24	18	030	CAN (cancel)	56	38	070	8	8	88	58	130	X	X	120	78	170	x	x
25	19	031	EM (end of medium)	57	39	071	9	9	89	59	131	Y	Y	121	79	171	y	y
26	1A	032	SUB (substitute)	58	3A	072	:	:	90	5A	132	Z	Z	122	7A	172	z	z
27	1B	033	ESC (escape)	59	3B	073	;	;	91	5B	133	[	[	123	7B	173	{	{
28	1C	034	FS (file separator)	60	3C	074	<	<	92	5C	134	\	\	124	7C	174	|	
29	1D	035	GS (group separator)	61	3D	075	=	=	93	5D	135	]	]	125	7D	175	}	}
30	1E	036	RS (record separator)	62	3E	076	>	>	94	5E	136	^	^	126	7E	176	~	~
31	1F	037	US (unit separator)	63	3F	077	?	?	95	5F	137	_	_	127	7F	177		DEL

Phred scores are ASCII encoded

- Add an offset and convert the sum to ASCII
- **Sanger / Illumina 1.8+ encoding**
 - Offset: 33; Phred score range: 0-41;
 - Q41 → chr(74) → J
 - used by all current sequencers
- This encoding would mean that there are 42 different values
- But the FASTQ no longer contains 42 distinct values.
- NovaSeq X with XLEAP-SBS reports a maximum of Q40, binned into three values (Q12 / Q24 / Q40)
- over 90% of bases fall in the top bin — per-base quality is close to a 2-bit variable

Table 1: Q-Score Bins for an Optimized 8-Level Mapping

Quality Score Bins	Example of Empirically Mapped Quality Scores*
N (no call)	N (no call)
2–9	6
10–19	15
20–24	22
25–29	27
30–34	33
35–39	37
≥ 40	40

Q30 Is a Per-Base Number — Reads Are Longer Than One Base

$P(\text{read contains } \geq 1 \text{ error}) = 1 - (1 - e)^L$ — with e the per-base error and L the read length



→ at Q30, a 150 bp read has a 14% chance of carrying at least one wrong base; at Q20 it is 78%

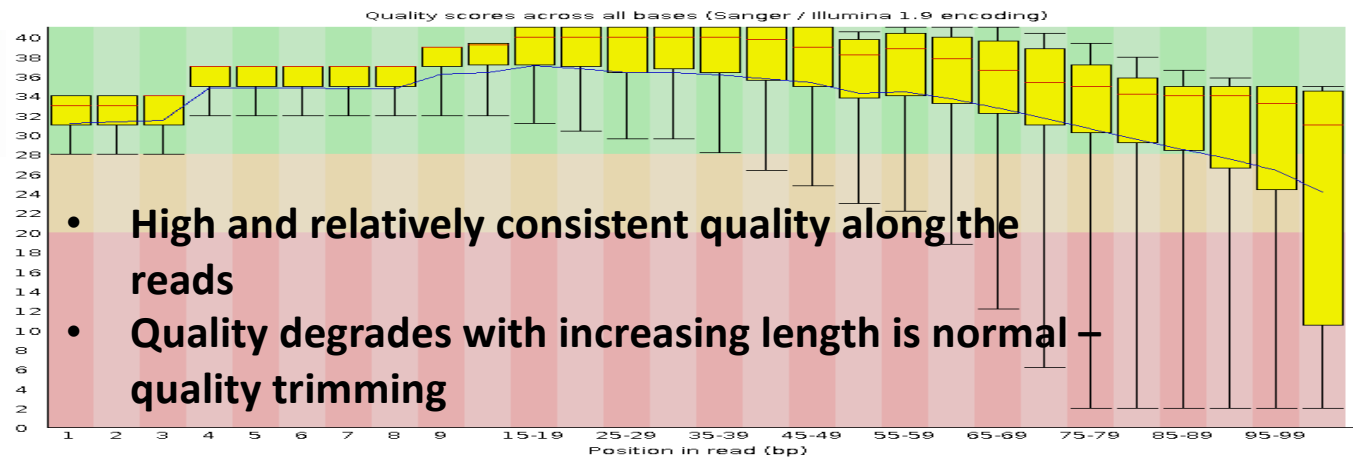
Read Quality Control

- Library construction could introduce bias
 - Fragmentation, ligation, amplification
 - GC bias
 - Over-amplification
 - Contamination
- Sequencing errors:
 - Chemical, optical, computational
- Type of errors:
 - random: more reads average out the errors
 - systematic: coverage does not help

Platform	Dominant error	Raw read accuracy	Consensus / duplex
Illumina SBS	substitution	~0.1% (Q30+)	—
PacBio Revio	indel	HiFi only: Q30–Q40	—
Nanopore R10.4.1	indel	up to 99.75% (Q26)	duplex > Q30

Per base sequence quality - FastQC

- Range of quality values across all bases at each position

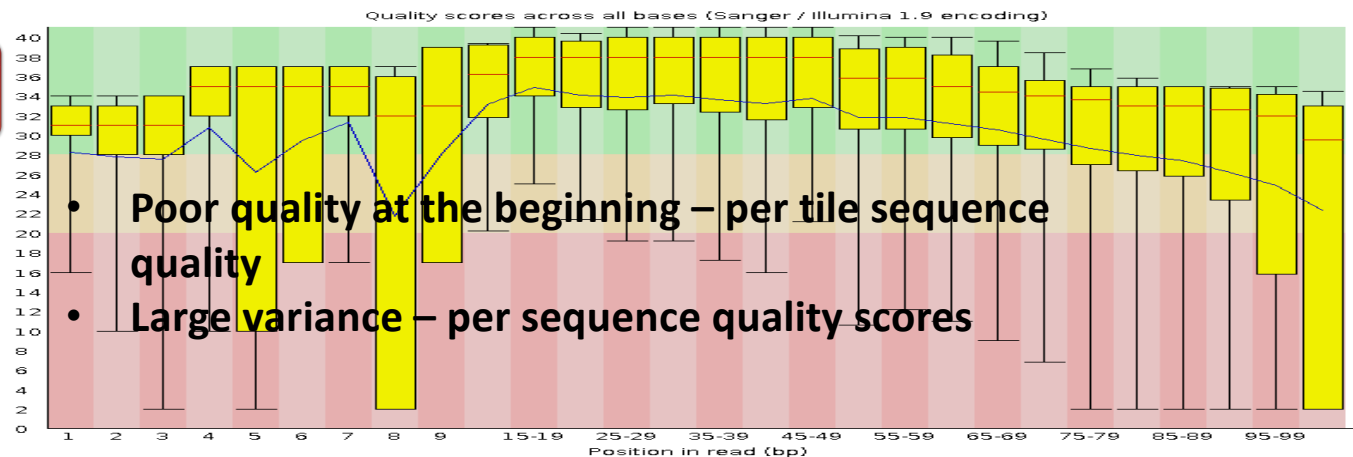


Green: >Q28, good

Orange: >Q20, reasonable

Red: <Q20, poor

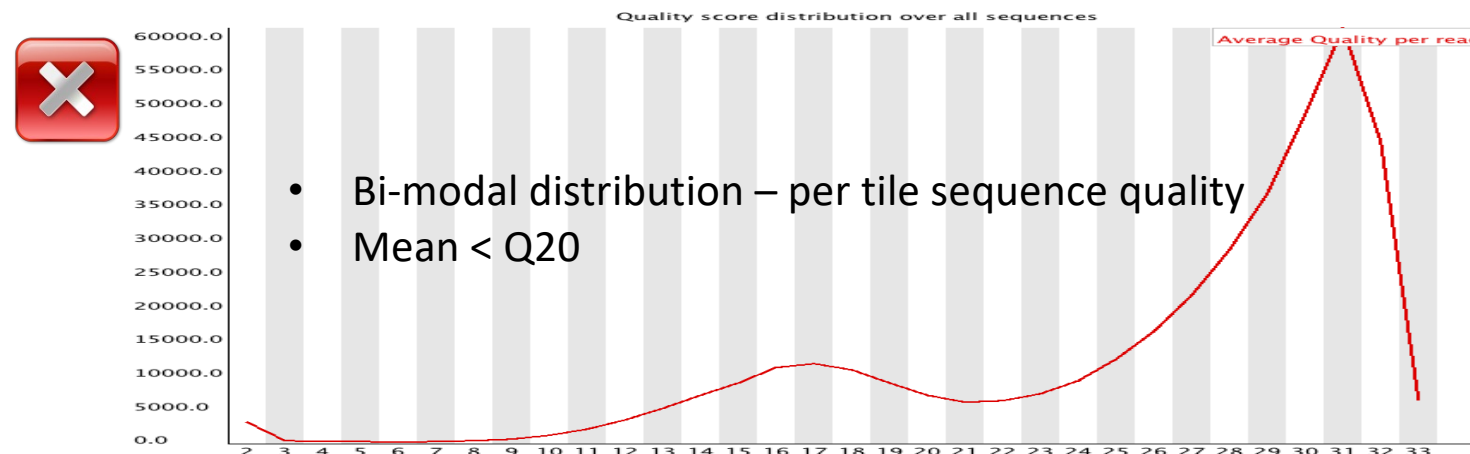
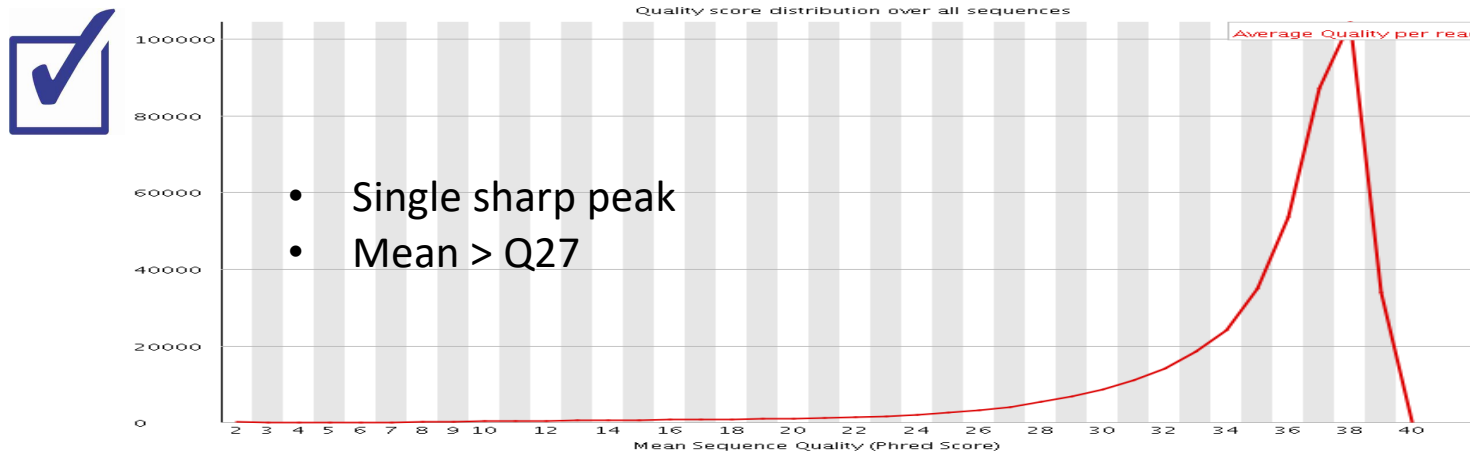
Median > Q25



Median < Q20

Per sequence quality scores - FastQC

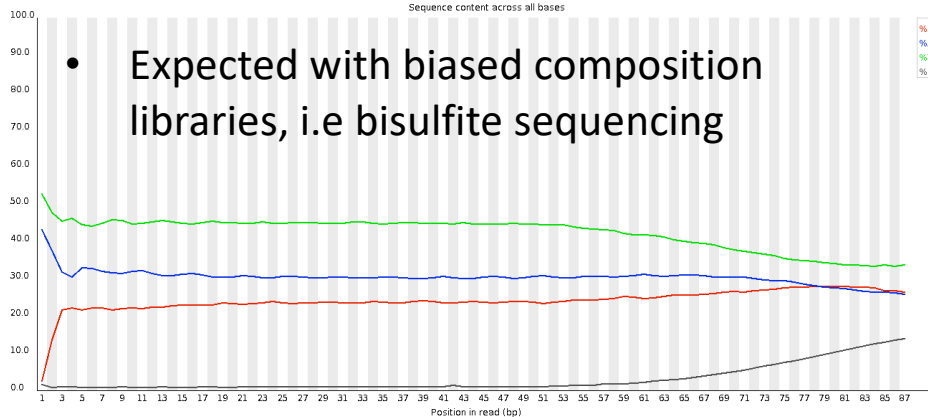
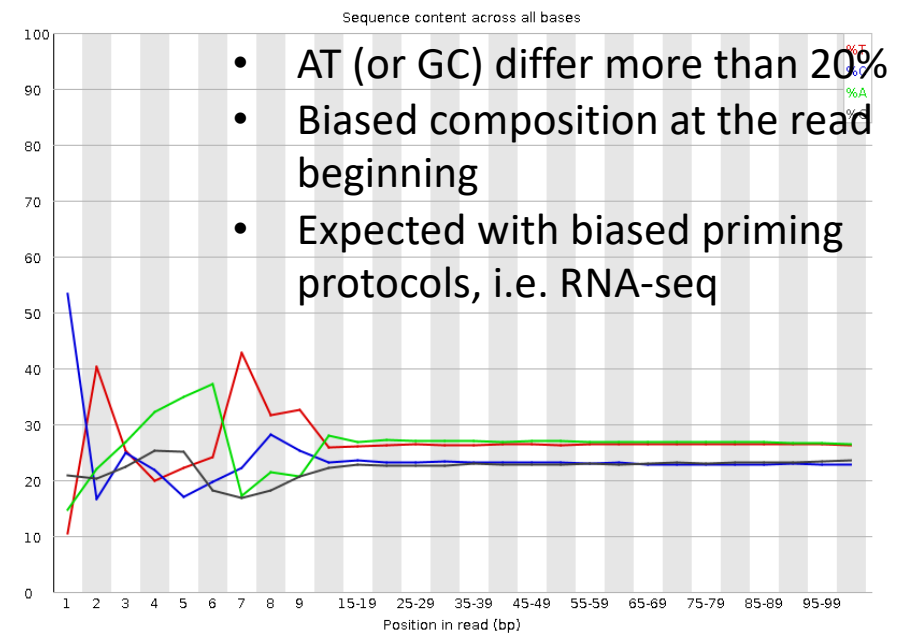
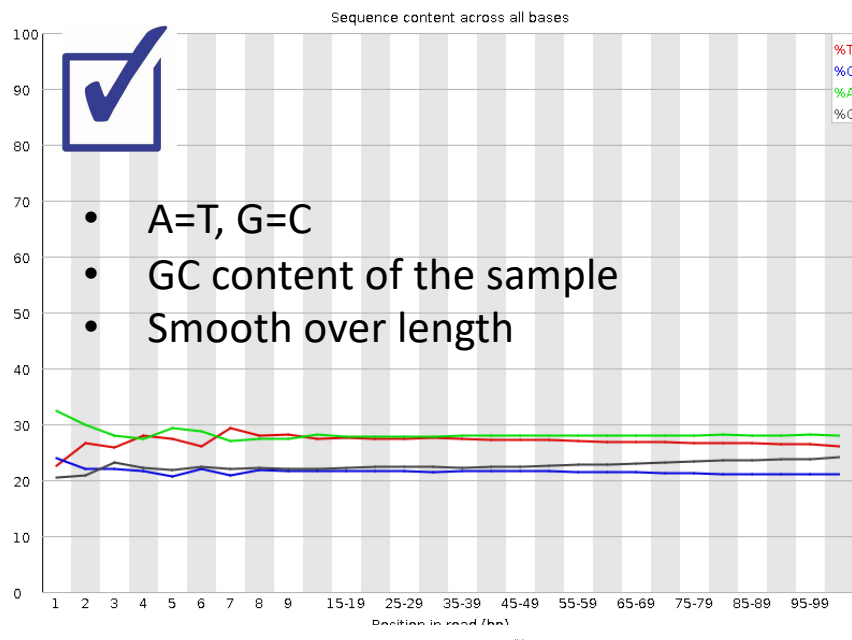
- Subset of sequences with universally low quality values



Failure: $< \text{average} - 5$

Per base sequence content - FastQC

- The portion of A, T, G, and C at each position



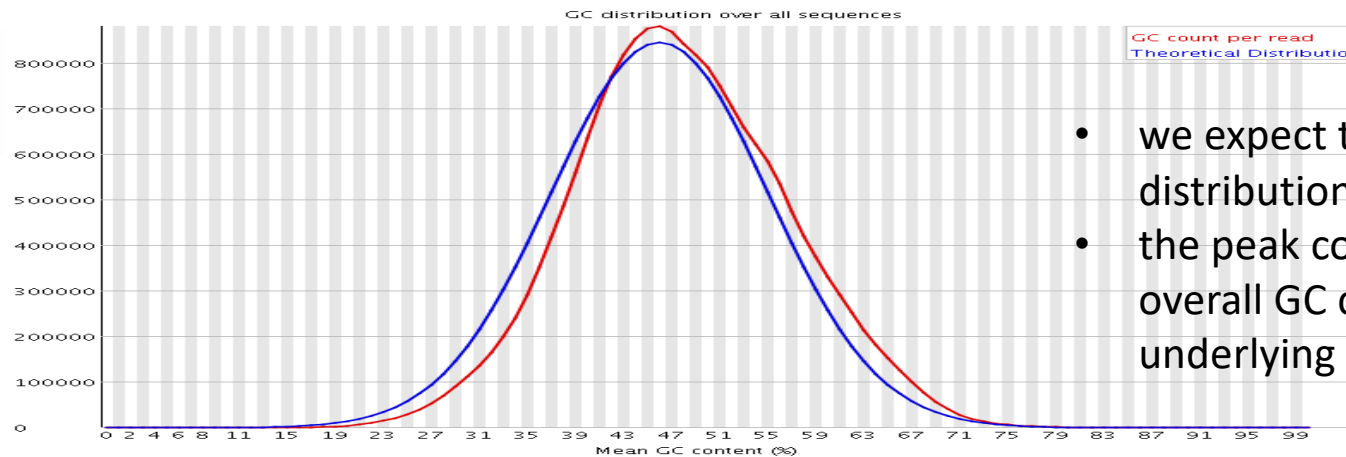
Biases in Illumina transcriptome sequencing caused by random hexamer priming

Kasper D. Hansen^{1,*}, Steven E. Brenner² and Sandrine Dudoit^{1,3}

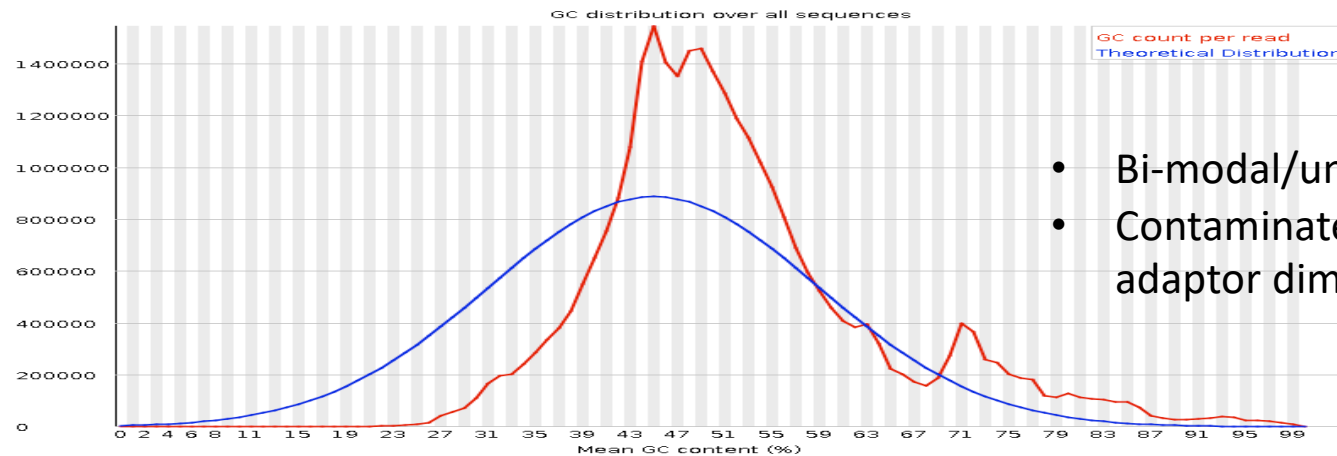
Treatment of DNA with bisulfite converts cytosine to uracil, but leaves methylated cytosine unaffected. Therefore, DNA that has been treated with bisulfite retains only methylated cytosines.

Per sequence GC content - FastQC

- Distribution of average GC in all reads



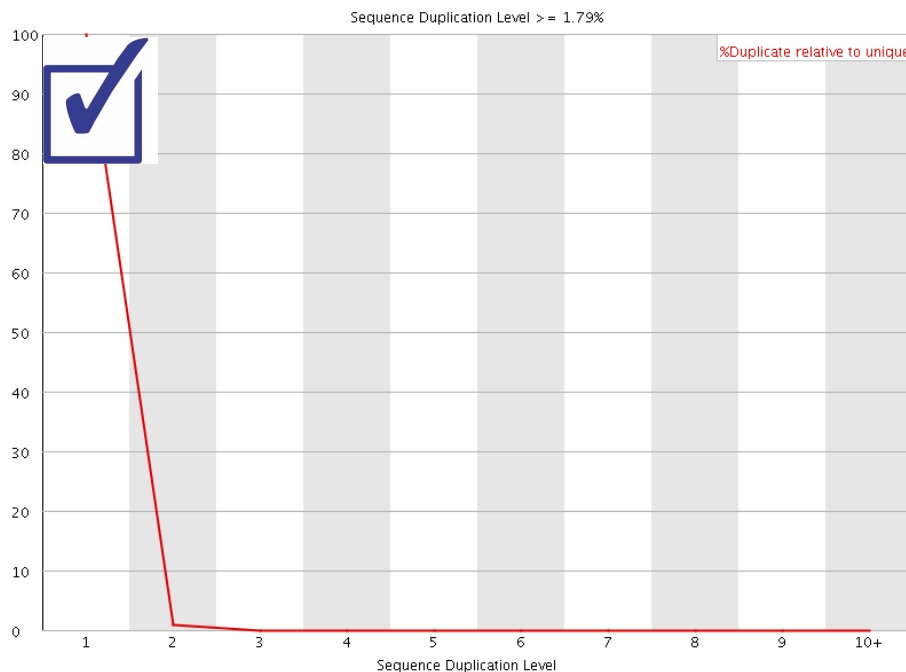
- we expect to see a roughly normal distribution of GC content
- the peak corresponds to the overall GC content of the underlying genome



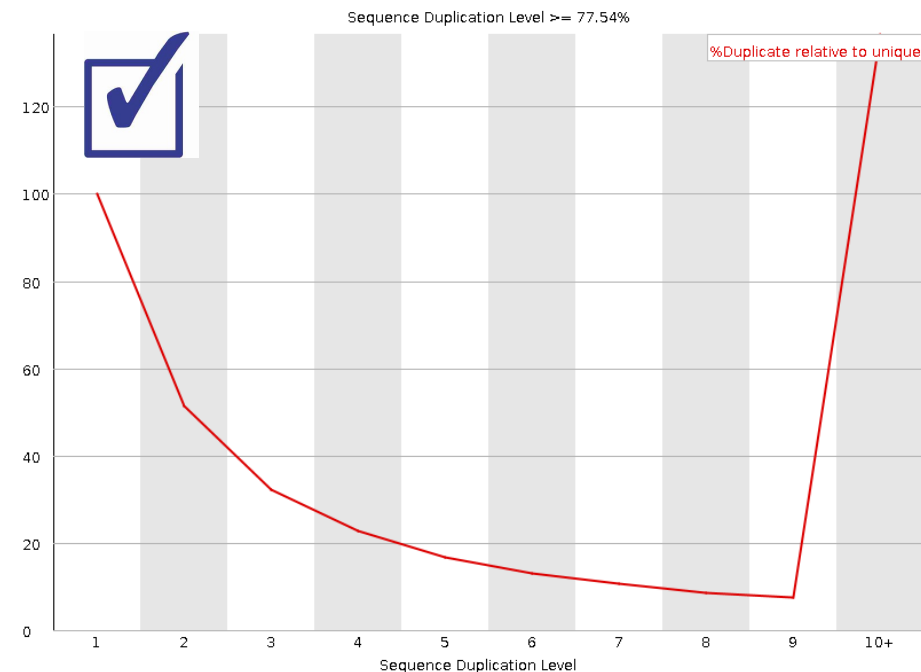
- Bi-modal/unusual distribution
- Contaminated/biased subset, i.e. adaptor dimmers, rRNA etc

Sequence duplication - FastQC

- Relative number of sequences with different degrees of duplication



- Essentially no duplication



High duplication levels:

- DNA-seq: PCR over amplification, too little input material
- Normal in RNA-seq: high expression

Overrepresented sequences - FastQC

- Sequences make up >0.1 % of the total
- Compare those with a contamination database for finding contamination (i.e. adaptor dimmers)

Overrepresented sequences

Sequence	Count	Percentage	Possible Source
GGAAGAGCACACGTCTGAACTCCAGTCACCAAGATCATCTCGTATGCCGTC	75874	1.5613887498682963	TruSeq Adapter, Index 7 (100% over 50bp)
GGAAGAGCACACGTCTGAACTCCAGTCACCGATGTATCTCGTATGCCGTC	7636	0.15713900010536297	TruSeq Adapter, Index 2 (100% over 50bp)
GGAAGAGCACACGTCTGAACTCCAGTCACACAGTGATCTCGTATGCCGTC	7539	0.1551428656095248	TruSeq Adapter, Index 5 (100% over 50bp)
GGAAGAGCACACGTCTGAACTCCAGTCACGCCAATATCTCGTATGCCGTC	5117	0.10530123933199874	TruSeq Adapter, Index 6 (100% over 50bp)

- Can be normal and biologically meaningful
 - highly expressed transcripts
 - high copy number repeats
 - Less diverse library (amplicons)

A Red Cross Is Not a Verdict

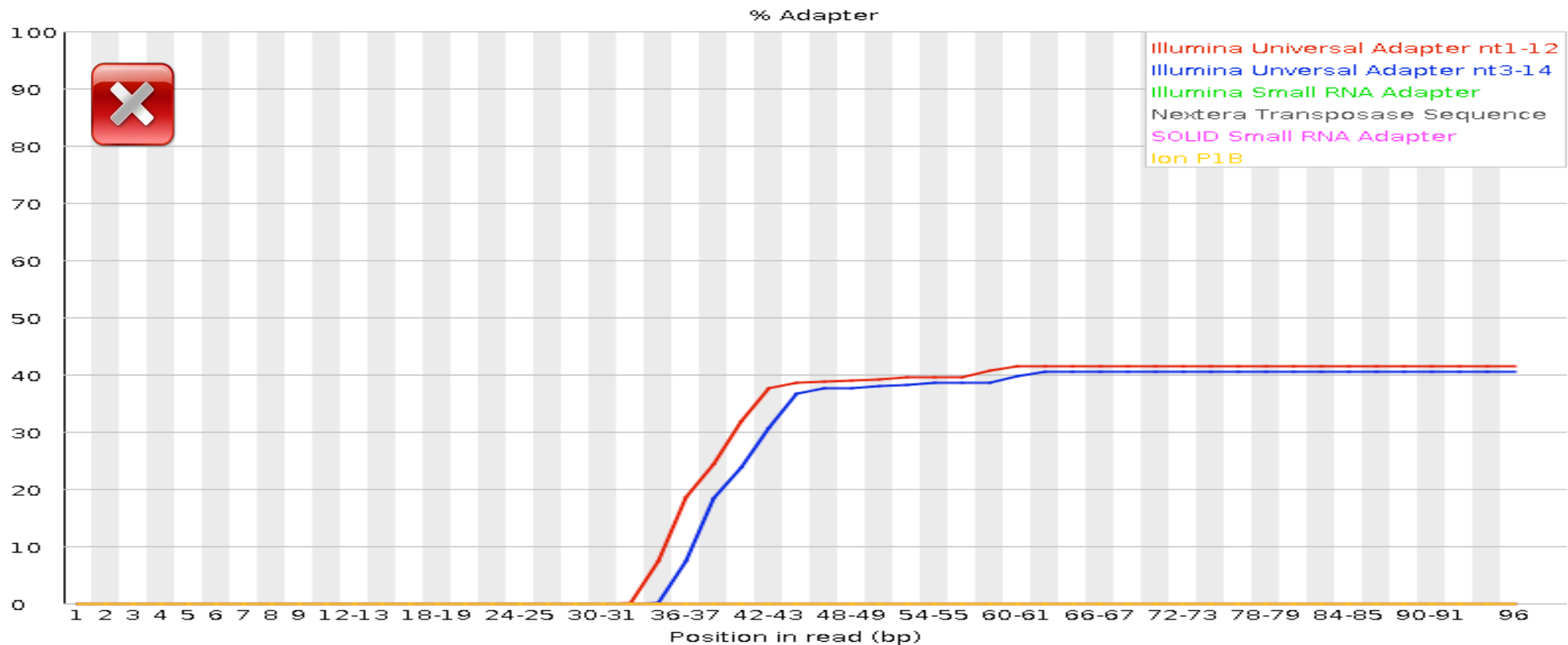
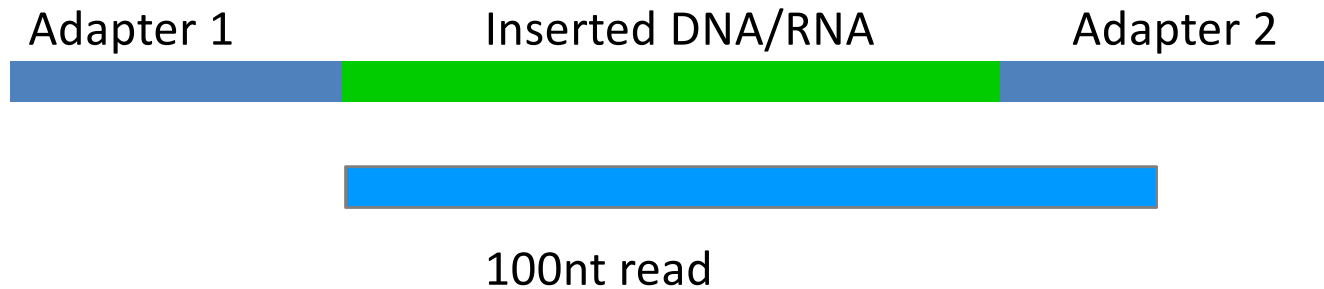
FastQC's thresholds are fixed heuristics, calibrated on whole-genome DNA sequencing of a random library.

Modules that are **expected** to fail, by assay:

- **RNA-seq** — per-base sequence content (random-hexamer priming), duplication (highly expressed genes), overrepresented sequences
- **amplicon / 16S** — sequence content, GC distribution, duplication: the library is deliberately not diverse
- **bisulfite** — sequence content, by construction: most C has become T
- **single-cell** — duplication and overrepresentation, because of amplification and the barcode structure

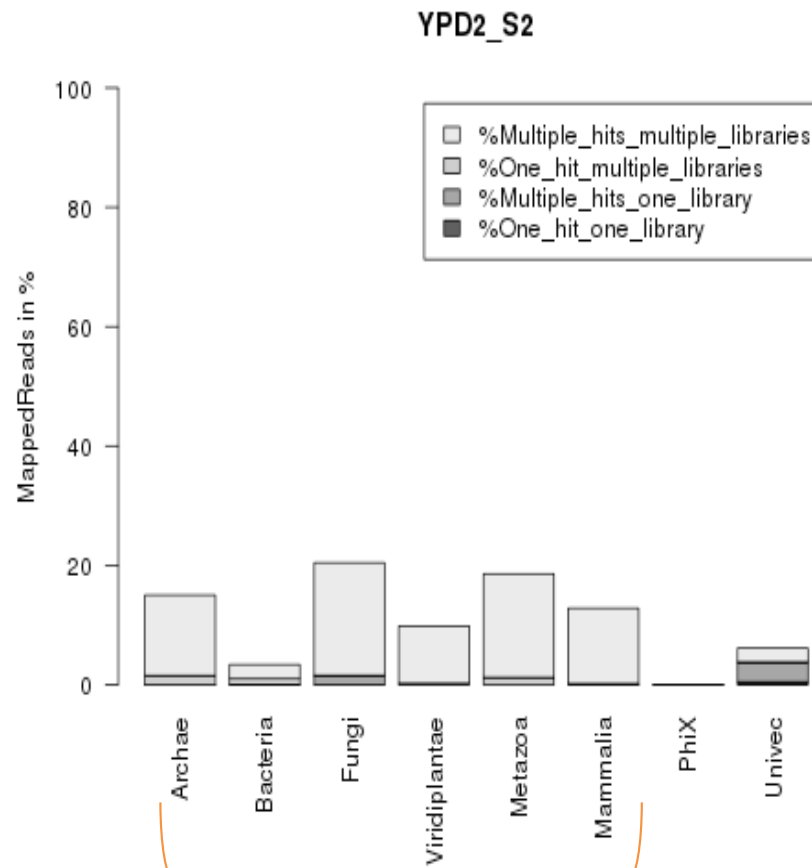
→ understand what sample and library prep was done and interpret the fastqc output correspondingly

Adapter Content - FastQC

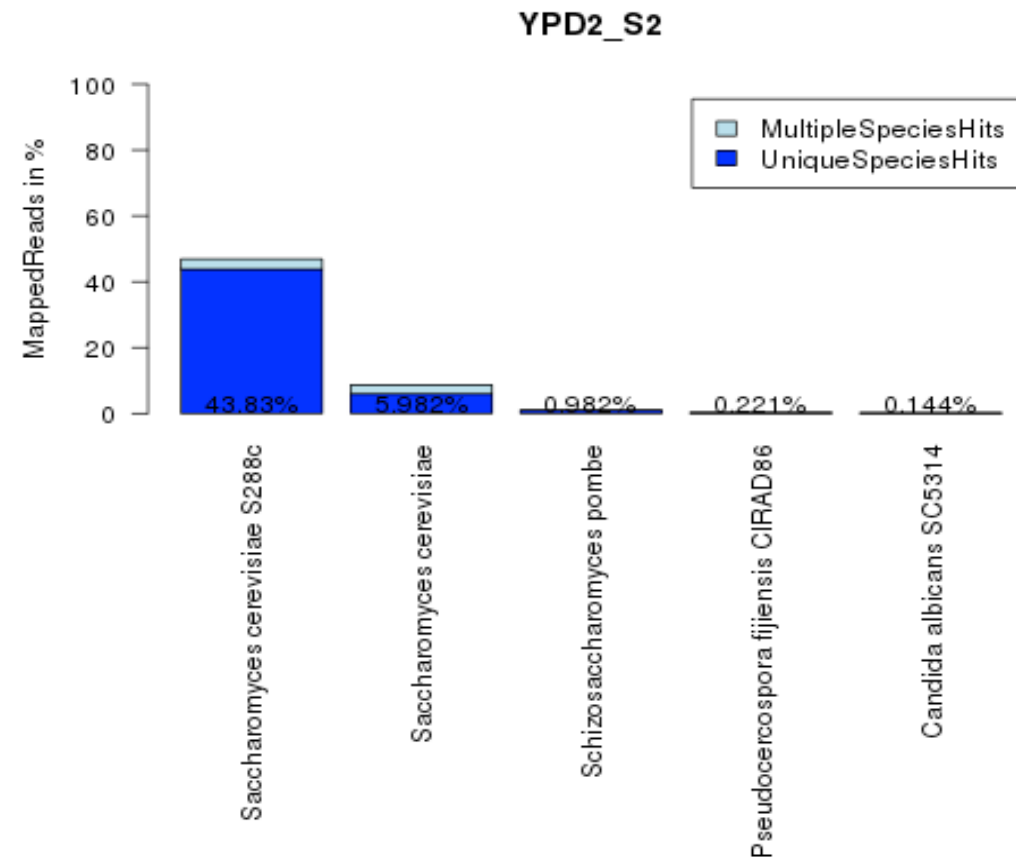


- How accurate was the sequencing → Fastqc
- Are these reads the intended ones → FastqScreen

Contamination Check - FastqScreen



rRNA-Content (Silva)



Mapping to RefSeq mRNAs
(all species)

Data preprocessing common tasks

1. Trimming: remove bad bases from the ends of the reads

- Adapter sequence
- Low quality bases

2. Filtering: remove bad reads

- Low quality reads
- Contaminating sequences
- Low complexity reads (repeats)
- Short (<20bp) reads – they slow down mapping software

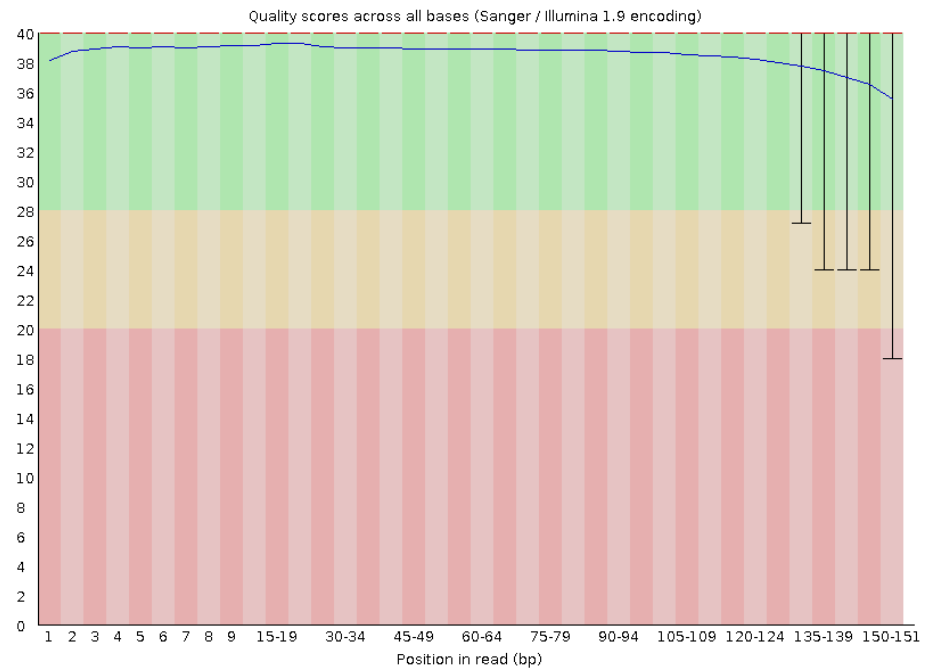
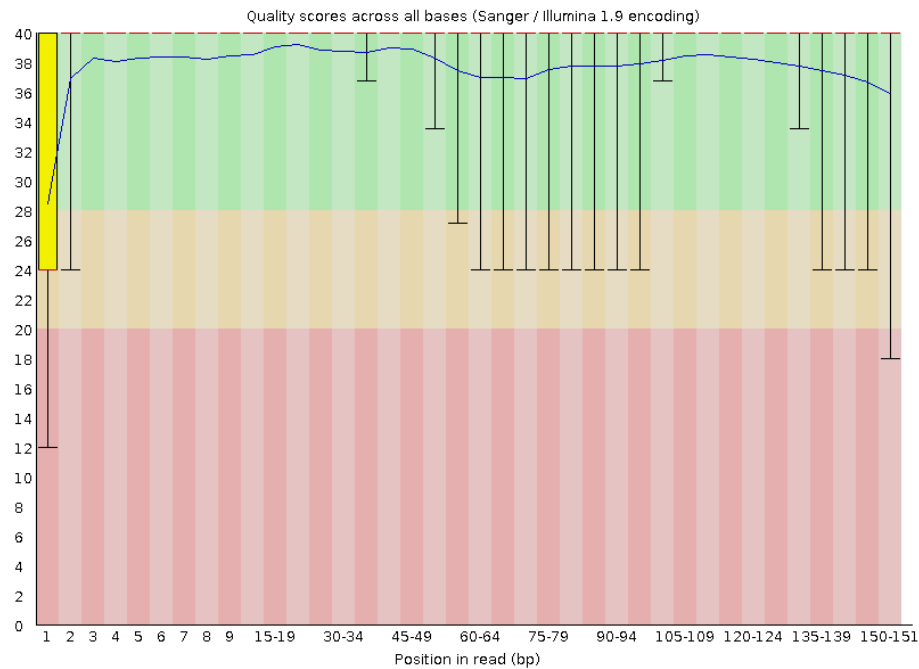
Data preprocessing software

- fastp
 - <https://github.com/OpenGenome/fastp>
 - Adapter trimming, quality trimming & filtering, ...
- Trimmomatic
 - <https://github.com/usadellab/Trimmomatic>
 - Adapter trimming, quality trimming & filtering, ...
- FlexBar (FAR)
 - <https://github.com/seqan/flexbar>
 - Flexible barcode detection and adapter removal
- cutadapt
 - <https://cutadapt.readthedocs.io/en/v4.0/index.html>
 - remove adaptors and types of wanted sequence
- TagCleaner
 - <http://tagcleaner.sourceforge.net>
 - Trim MIDs or adaptors, demultiplexing
- DeconSeq
 - <http://deconseq.sourceforge.net>
 - Remove potential contaminants

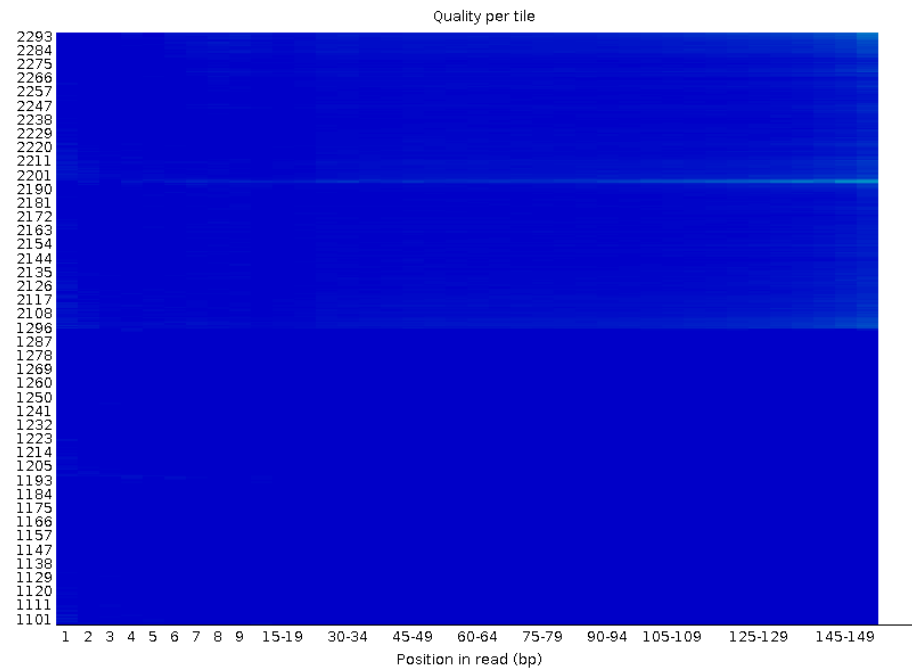
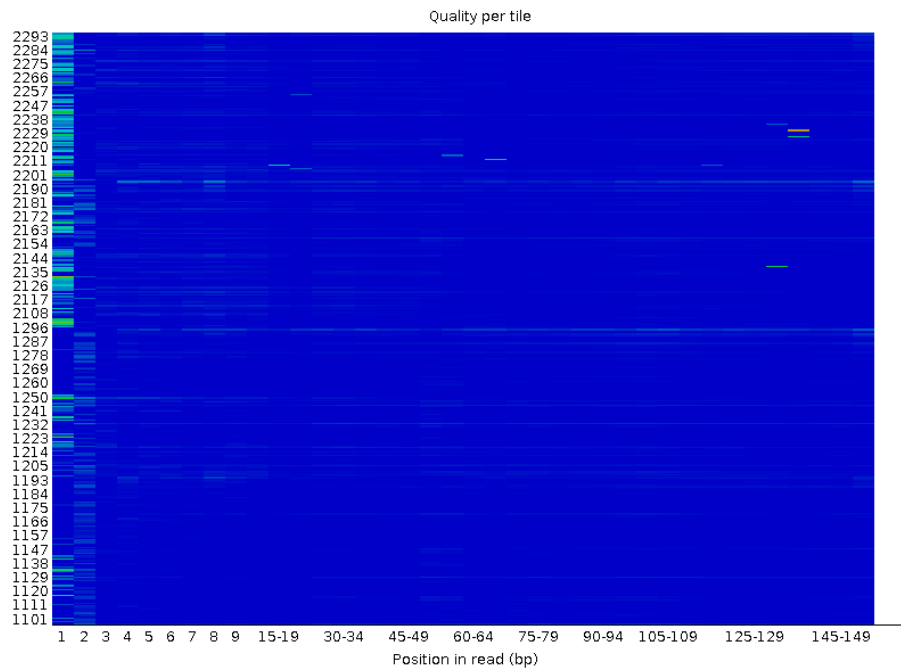
Example: Batch effect

- On the following slides: QC plots from two samples done in two batches
- The RNA-seq samples were prepared in two batches, and the difference is visible in several FastQC modules
- In this dataset the gene-level estimates were unaffected (checked by replicates)

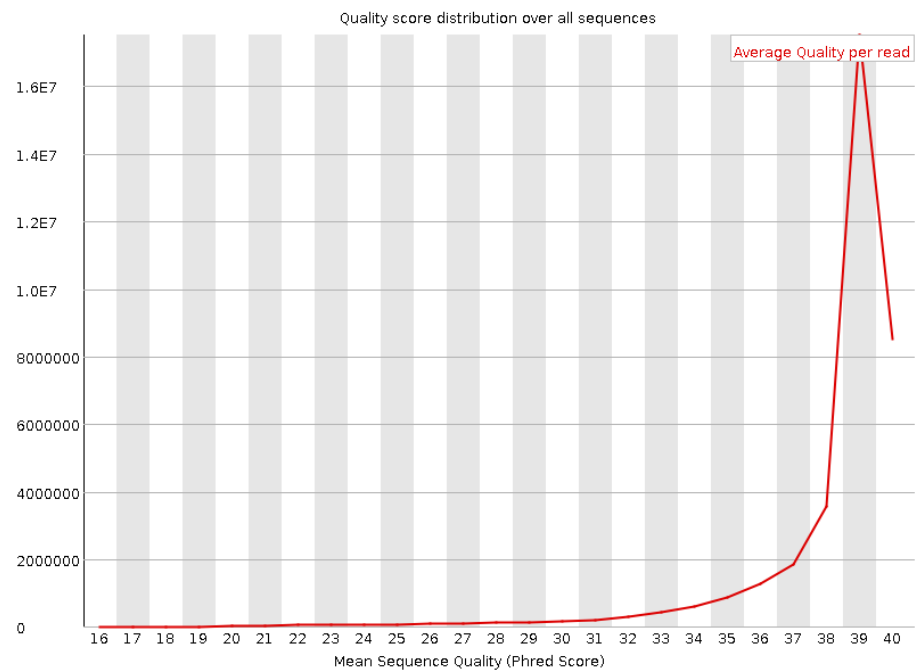
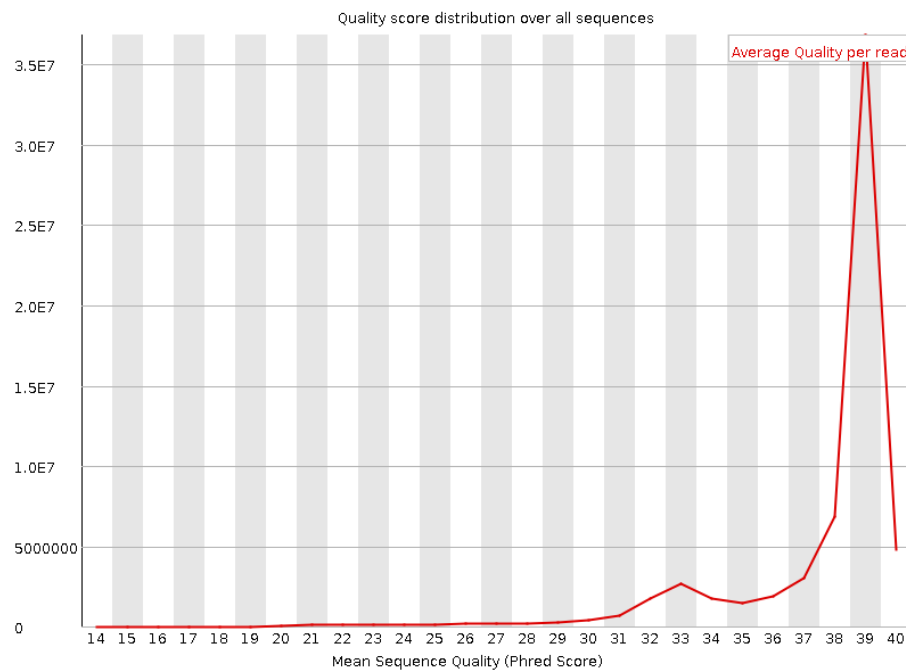
Per-base quality



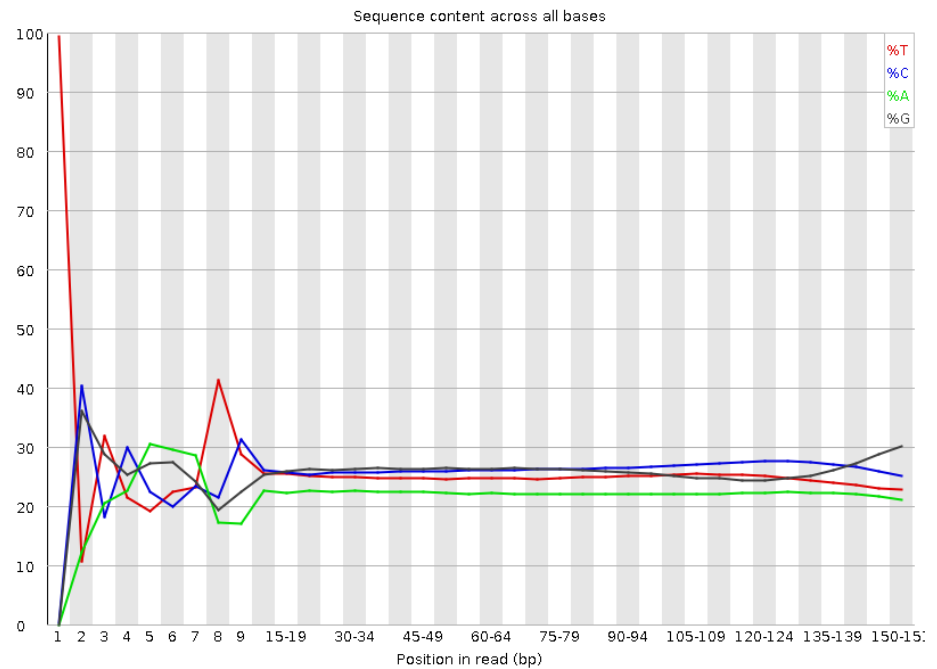
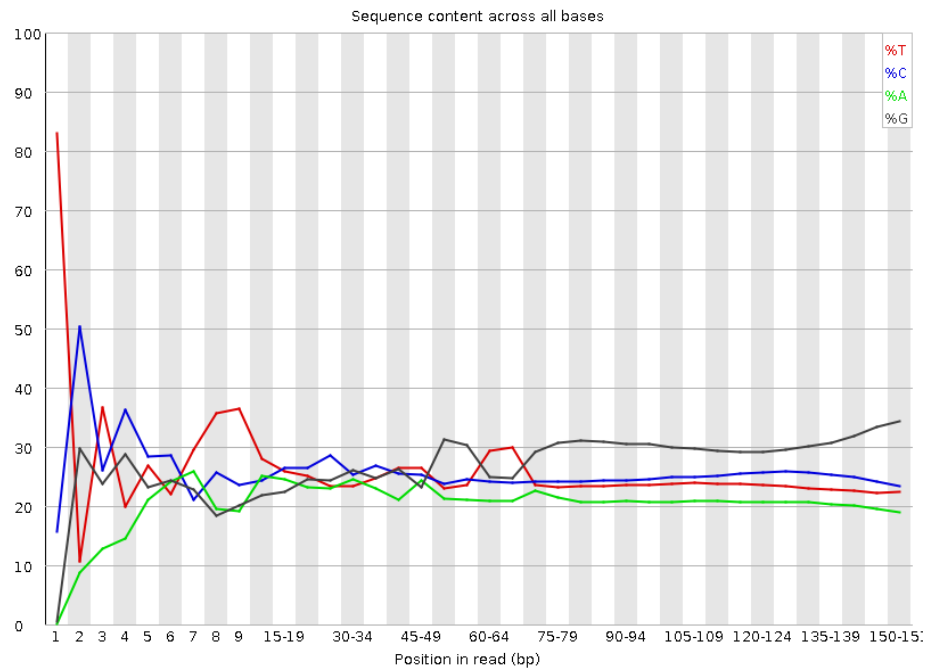
Per tile sequence quality



Per Sequence Quality



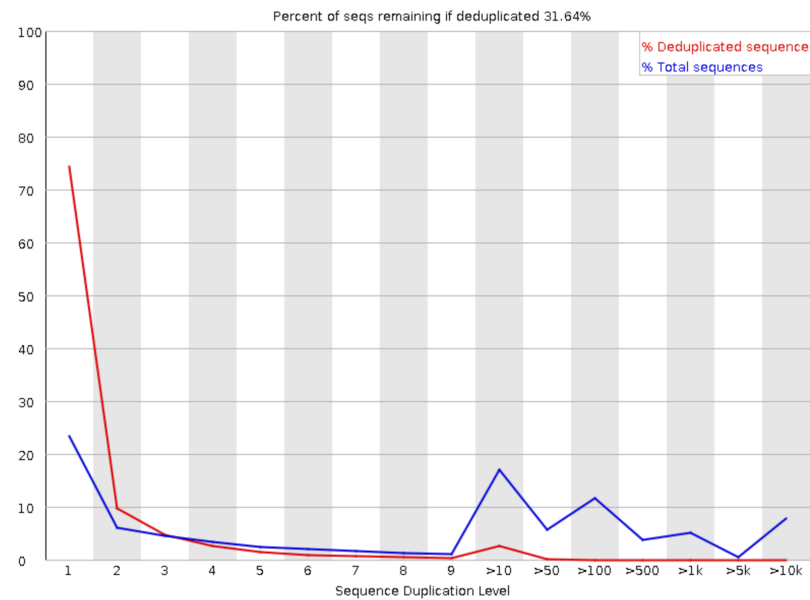
Per-base sequence content



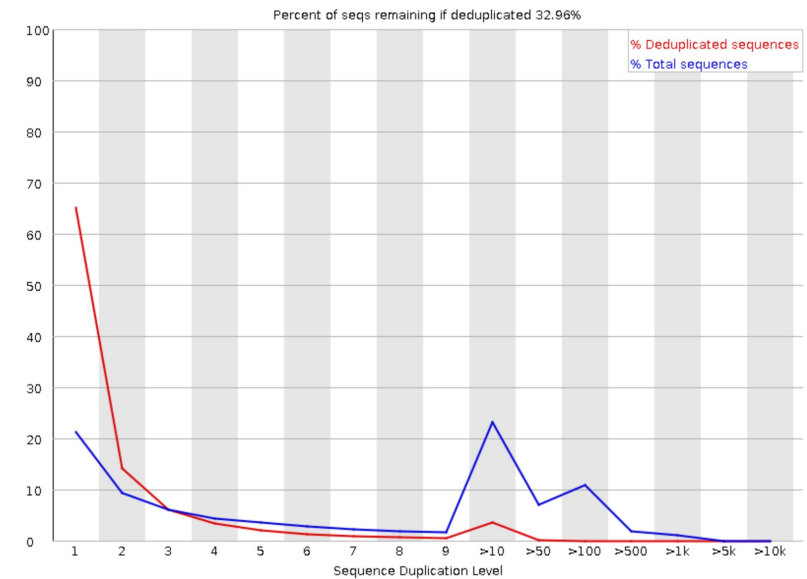
- illumina RNA ligation protocol: Adds a T in front of every read

Duplication Levels

Sequence Duplication Levels



Sequence Duplication Levels



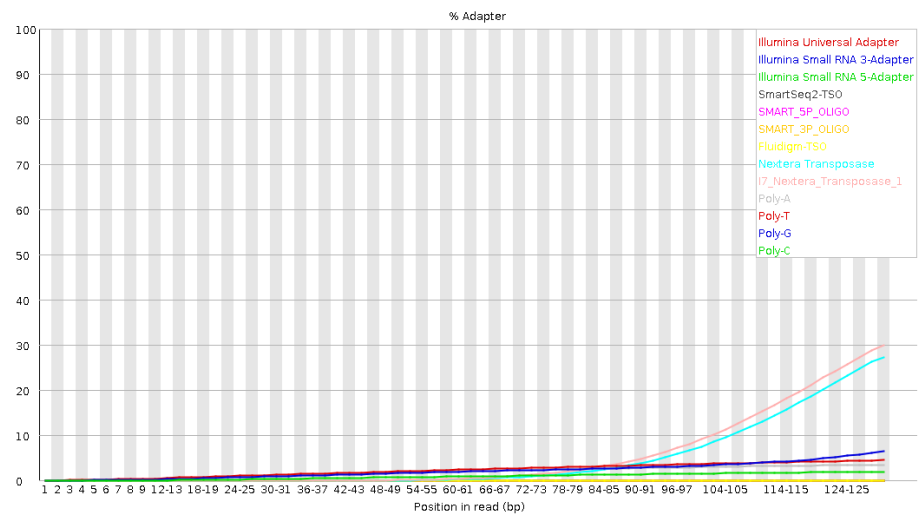
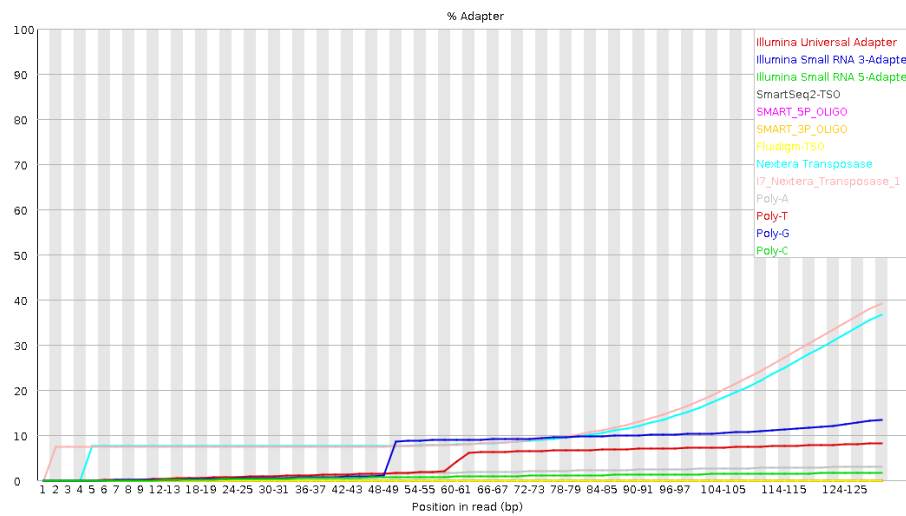
Overrepresented sequences

Sequence	Count	Percentage	Possible Source
TCTGTCTCTTATACACATCTCCGAGCCACGAGACCGATCTGTGAATCTG	3679233	5.668340356813218	No Hit
CCTGTCTCTTATACACATCTCCGAGCCACGAGACCGATCTGTGAATCTG	674594	1.0393004179577796	No Hit
TCTGTCTCTTATACACATCTCCGAGCCACGAGACCGATCTGTGAATCTA	182059	0.28048573629913015	No Hit

Overrepresented sequences

No overrepresented sequences

Adapter Content



Recommendations

- Always generate quality control plots visualizing key characteristics for all libraries
- Trim and/or filter data if needed
- Applications where erroneous reads are of concern:
 - de novo assembly
 - low coverage variant calling
- Applications that are more tolerant to low quality bases
 - RNA-seq
 - ...

and before any of it: understand the protocol. The prep decided how these reads were constructed and how they should be interpreted